English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Mathematics / Applied Mathematics, 7 June 2025 Test 2 ___*The exact grading formula will depend on the results. As an approximation, use 2 + the number of points. Time allowed: 4 hours. Good luck!*

Problem 1. (0.5 points) Information displays with 10×100 diodes are installed at public transport stops. Model their lifetimes as iid random variables with mean 10 years. Experience shows that when fewer than 75% of the diodes work, the information is often unreadable and the display needs repair. Determine a condition under which the probability of needing repair after k years is approximately 90%, for $k = 1$ or 5.

Problem 2. (0.25 points) Let ξ, η be independent, $\xi \sim \text{Exp}(2)$ and $\eta \sim U(0,3)$, so

$$f_\xi(x) = \begin{cases} 2e^{-2x}, & x > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad f_\eta(x) = \begin{cases} 1/3, & 0 < x < 3, \\ 0, & \text{otherwise}. \end{cases}$$

Find their correlation, $\mathbb{P}(\xi < \eta)$ and the density of ξ/η.

Problem 3.

1. (0.25 points) Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, with a sample of n observations. Test $H_0 : \mu = 3.8$ against $H_1 : \mu \neq 3.8$ at significance level $\alpha = 0.05$. Can H_0 be rejected for $n = 16$, $\overline{X} = 3.295$, $S^2 = 1.4641$? Derive the test using likelihood ratios.

2. (0.5 points) Let $X \sim U(-\alpha, \alpha)$. Find method of moments and maximum likelihood estimators of α. Determine whether they are unbiased and/or consistent.

Problem 4. Let $X_1, X_2, \ldots$ be identically distributed, and $\overline{X}_n = (X_1 + \cdots + X_n)/n$.

1. (0.25 points) If $X_i \sim \mathcal{N}(\mu, \sigma^2)$, determine the distribution of $\overline{X}_n$. Calculate the moment generating function of X_i and use it to find $\mathbb{E}[X_i^k]$ for $k \leq 4$.

2. (0.25 points) State the law of large numbers.

3. (0.25 points) Can $D\overline{X}_n$ fail to tend to 0 as $n \to \infty$? Can the X_i take finitely many values while $\overline{X}_n$ fails to converge to a constant?

4. (0.5 points) If all X_i have density $f_X(x) = 1/[\pi(1 + x^2)]$ for $x \in \mathbb{R}$, find the moment generating function $M_X(t) = \mathbb{E}[e^{tX_1}]$ and characteristic function $\varphi_X(t) = \mathbb{E}[e^{itX_1}]$. Calculate $\mathbb{E}[X_1]$ using M_X. Determine the distribution of $\overline{X}_n$ by a suitable theorem or calculation.

Problem 5. Let $X_1, \ldots, X_n$ be iid with density

$$f_X(x) = \begin{cases} \alpha/(1 + x)^{\alpha+1}, & x > 0, \\ 0, & \text{otherwise}, \end{cases} \qquad \alpha > 0.$$

Let $X_{(1)} \leq \cdots \leq X_{(n)}$ be the order statistics and $Y_k = \ln X_{(k)}$.

1. (0.25 points) Calculate $\mathbb{E}[X_i^k]$ for integer k. Find the density of $X_{(i)}$ and a closed expression for $\mathbb{E}[X_{(1)}]$.

2. (0.5 points) Find explicit expressions for $\mathbb{E}[Y_k]$ and $D(Y_k)$. For which k is $D(Y_k)$ minimized?

Problem 6. A vector $X = (X_1, \ldots, X_n)$ is normally distributed, or Gaussian, with mean $\mu = (\mu_1, \ldots, \mu_n)$ and covariance matrix $\Sigma_{ij} = \text{Cov}(X_i, X_j)$ if every linear combination $\langle \lambda, X \rangle = \sum_i \lambda_i X_i$ has distribution $N(\langle \lambda, \mu \rangle, \sum_{i,j} \lambda_i \lambda_j \text{Cov}(X_i, X_j))$.[1]

1. (0.25 points) Find $M_Y(t)$ for $Y \sim N(\mu, \sigma^2)$. Using the fact that $M_Y(t_1) M_W(t_2) = M_{Y,W}(t_1, t_2) = \mathbb{E}[e^{t_1 Y + t_2 W}]$ in a nonempty neighbourhood of 0 implies independence, and the analogous result for more than two variables, prove that uncorrelated components of a Gaussian vector are jointly independent normally distributed variables.

   For the rest of the problem, assume X consists of independent $N(0,1)$ variables.

2. (0.25 points) Let P be an $n \times n$ orthogonal matrix, $PP^T = I_n$. Prove that PX also consists of independent $N(0,1)$ variables.

3. (0.25 points) Let V be a k-dimensional linear subspace of $\mathbb{R}^n$ and $V^\perp$ its orthogonal complement. Prove that the projections of X onto these subspaces are independent Gaussian vectors, and that their lengths are independent and chi-squared distributed with parameters k and $n - k$, respectively.

   Using the preceding part, explain why it suffices to consider $X_V = (X_1, \ldots, X_k, 0, \ldots, 0)$ and $X - X_V$.

[1] The sum is the quadratic form $\lambda \Sigma \lambda^T$.

4. (0.25 points) Take V to be spanned by $(1, \ldots, 1)$. Prove that

$$\overline{X} = \frac{X_1 + \cdots + X_n}{n} \qquad \text{and} \qquad (n-1)S^2 = \sum_{i=1}^{n} (X_i - \overline{X}_n)^2$$

are independent, and specify their distributions.

Problem 7. *Bonus* (0.25 points) Find $\lim_{n\to\infty} e^{-n}\left(\frac{n}{1!} + \frac{n^2}{2!} + \cdots + \frac{n^n}{n!}\right)$.